

You-Ting Wu (he/him)

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Research Interests/Keywords

climate modeling, Arctic climate change, machine learning in climate change, energetic feedbacks, climate risk

Education

Sept 2024–present	Columbia University , New York, NY, USA Ph.D., <i>Applied Mathematics and Climate Science</i> — Advisors: Lorenzo Polvani and Michael Previdi
Sept 2018 -July 2023	National Taiwan University , Taipei, Taiwan Double BSc., <i>Mechanical Engineering</i> and <i>Atmospheric Sciences</i>

Publications

- Previdi, M., **Wu, Y.-T.**, Polvani, L. M., England, M. R., Sigmond, M., and Liang, Y.-C. Substantial twentieth-century global sea level rise caused by ozone-depleting substances. Submitted to *Journal of Climate* (under review).
- Wu, Y.-T.**, Previdi, M., Romanou, A., Lerner, P., and Polvani, L.M. Arctic Climate Tipping Points under Delayed CO2 Emissions Mitigation. Accepted by *Communications Earth & Environment* (to appear).
- Liang, Y.-C., Polvani, L. M., Previdi, M., Dong, Y., **Wu, Y.-T.**, England, M.R., and Griffith, S. M. (2026). The importance of non-CO2 greenhouse gases to Arctic warming and sea-ice loss. *Geophysical Research Letters*, 53, e2026GL122867.
- Wu, Y.-T.**, Liang, Y.-C., Previdi, M., Polvani, L. M., England, M. R., Sigmond, M., and Lo, M.-H. (2024). Stronger Arctic Amplification from Anthropogenic Aerosols than from Greenhouse Gases. *npj Climate and Atmospheric Science* 7 (1), 142.
- Wu, Y.-T.**, Liang, Y.-C., Kuo Y.-N., Lehner, F., Polvani, L. M., Previdi, M., Lo, M.-H., Lan, C.-W. (2023). Exploiting SMILES and the CMIP5 archive to understand Arctic climate change seasonality and uncertainty. *Geophysical Research Letters*, 50(2), e2022GL100745.

Research Experiences

Sept. 2024–present	Graduate Research Assistant , Columbia University <i>New York, NY, USA</i> Quantifying and understanding the impacts of various anthropogenic emissions on future Arctic climate system using observations, climate models and data-driven statistical methods
June 2021–July 2024	Undergraduate Researcher , Polar Climate Research Group, National Taiwan University <i>Taipei, Taiwan</i> - Implemented physics-based climate models to identify the impacts of different anthropogenic forcings on climate change - Processed geospatial data to quantify key climate indices; translated complex data into accessible visuals to understand projected Arctic climate
June 2022–Aug 2022	Visiting Student , Lamont-Doherty Earth Observatory, Columbia University <i>Palisades, NY, USA</i> Utilized new climate models to analyze Arctic responses to aerosol emissions

Awards and Honors

Jun 2026	Government Scholarship to Study Abroad (US\$32,000) <i>Ministry of Education, Taiwan</i>
Jan 2026	Nominated for Membership of $\Sigma\Xi$ (The Scientific Research Honor Society) <i>USA</i>
June 2023	Excellent Undergraduate Research Project Award <i>Ministry of Science and Technology, Taiwan</i>
June 2022	Undergraduate Research Project Grants (US\$ 2,200) <i>Ministry of Science and Technology, Taiwan</i>

Presentations

Dec 2025	(poster) AGU25 Annual Meeting, New Orleans, LA <i>Arctic Climate Tipping Points under Delayed CO₂ Emissions Mitigation</i>
June 2025	(oral, invited) Polar Teatime Seminar, National Central University, Taoyuan, Taiwan <i>Stronger Arctic amplification from anthropogenic aerosols than from greenhouse gases</i>
Dec 2024	(oral) AMS 2025 Denver Summit, Denver, CO <i>Stronger Arctic amplification from anthropogenic aerosols than from greenhouse gases</i>
Dec 2024	(poster) AGU24 Annual Meeting, Washington, D.C. <i>Stronger Arctic amplification from anthropogenic aerosols than from greenhouse gases</i>
Oct 2022	(oral) Seminar at Earth and Life Science Department, University of Taipei (with Yu-Chiao Liang) <i>The Arctic Amplification, and its Seasonal Migration, under Greenhouse Gases Forcings.</i>
Aug 2022	(oral) AMS 2022 Collective Madison Meeting, Madison, WI <i>Decomposing Projection Uncertainty for the Seasonal Evolution of Arctic Amplification.</i>
Aug 2022	(oral) The 19 th Annual Meeting of Asia Oceania Geosciences Society, virtual <i>Decomposing Projection Uncertainty for the Seasonal Evolution of Arctic Amplification.</i>
June 2022	(oral) The 27 th Annual CESM Workshop, virtual <i>Decomposing Projection Uncertainty for the Seasonal Evolution of Arctic Amplification.</i>
Apr 2022	(oral) Seminar at Institute of Hydrological and Oceanic Sciences, National Central University, Taoyuan, Taiwan (with Yu-Chiao Liang) <i>The Arctic Amplification, and its Seasonal Migration, under Greenhouse Gases Forcings.</i>
Mar 2022	(oral) Seminar at Institute of Oceanography, NTU, Taipei, Taiwan (with Yu-Chiao Liang) <i>The Arctic Amplification, and its Seasonal Migration, under Greenhouse Gases Forcings.</i>
Sept 2021	(poster) Undergraduate Summer Research Program, National Taiwan University, Taipei, Taiwan <i>Seasonal Evolution and Uncertainty of Arctic Amplification in Multi-Model Large Ensembles.</i>
Aug 2021	(poster) 2021 Climate Hotpot in Action Forum, virtual. <i>Seasonal Evolution and Uncertainty of Arctic Amplification in Multi-Model Large Ensembles.</i>

Teaching Experiences

Jan 2025– June 2025	TA of Partial Differential Equations, Columbia University
Sept 2025– Dec 2025	TA of Introduction of Atmospheric Science, Columbia University
Jan 2025–June 2025	TA of Statistics, Data Analysis, and Coding for Sustainability Science, Columbia University
Sepe 2024–Dec 2024	TA of Physics of Fluids, Columbia University
Feb 2023–June 2023	TA of Numerical Analysis (twice), National Taiwan University
Feb 2022–June 2022	
Feb 2023–June 2023	TA of Workshop for Earth System Model, National Taiwan University
Feb 2023–June 2023	TA of Atmospheric Radiation, National Taiwan University
Sept 2022–Jan 2023	TA of Applied Mathematics I, National Taiwan University
Feb 2022–June 2022	TA of Middle-high Latitude Stratosphere-Troposphere Dynamics, National Taiwan University

Working Experiences

July 2023–Sept 2023	Software Engineering Intern , Formosa Climate Smart Service Limited <i>Taipei, Taiwan</i> – Built database infrastructure to convert engineering inputs into quantifiable Energy Use Intensity (EUI) metrics – Developed front-end platform to visualize building EUI reports – Designed a workflow to integrate geospatial data and engineering drawings into 3D building-EUI interfaces using AutoCAD, ArcGIS, and Python
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Skills

Computer Languages	Python, MATLAB, Fortran, GrADS, C/C++, HTML, Javascript
Computer-aided Design	Autodesk AutoCAD, Autodesk Inventor
Computational Fluid Analysis	ANSYS Fluent
Language	Mandarin (native), English (fluent), German (basic)